Ultrasonic measurements in PVC, polyacryl, aluminum, copper and water

I. Kaiser, J. Hölzle and S. Scheel

The sound velocities of water $v_{water} = (1.483 \pm 0.019)\ \frac{\mathrm{Km}}{\mathrm{s}}$, polyacrylic $v_{poly} = (2.657 \pm 0.055)\ \frac{\mathrm{Km}}{\mathrm{s}}$ and PVC $v_{PVC} = (2.302 \pm 0.023)\ \frac{\mathrm{Km}}{\mathrm{s}}$ were determined. All these values are within their error with the literature value. Furthermore, the speed of sound in copper and aluminum was determined as a function of the angle to the transmitting wave. These velocities correspond to the theoretically expected values. In PVC, the absorption coefficient was also determined, showing a frequency dependence. The higher the frequency, the stronger the absorption. It was also confirmed that higher frequencies have a better resolving power. A B-mode image was made of a polyacrylic block and with the same block it was confirmed that a precise length measurement can be made with the help of ultrasonic waves.

Executed: January 19, 2022
First submission: January 26, 2022

1 Ultrasound

Ultrasound examinations are an important part of modern life. They are always required when it comes to examining material that has a sensitive tissue structure. An example of this is the ultrasound examinations in pregnant women. An ultrasound scan can tell a lot about the child without putting the mother or the child at risk.

"Modern ultrasound systems do much more than produce images of unborn babies, however. Modern ultrasound systems are able to make detailed measurements of blood movements in blood vessels and tissues, visualize moving structures in 3D, and make measurements related to the stiffness of tissues" [3].

In this experiment, ultrasonic waves were used to determine the velocity of sound in different materials and the absorption coefficient in solids. Also, the dimensions of a polyacrylic brick were explored by examining reflection and transmission of ultrasound and a B-mode image was recorded.

2 Function of the ultrasonic echo scope

The ultrasonic echo scope uses a mechanical wave, its reflection / transmission and an echo program to examine a material. The Ultrasonic probes transform a high frequency electrical pulse into a sonic pulse. The sonic pulse travels as a sonic wave through the material. If the material has an inhomogeneity, the sound wave will be reflected by it. The reflected wave reaches the Ultrasonic probe and is transformed into an electrical signal.

This signal can be displayed in various graphical manners. Two graphical display methods were used in this experiment.

The first is the A-Mode (Amplitude Mode Scan). The reflected signal is displayed in a diagram. The x-axis displays the time the sonic wave traveled and the y-axis the intensity of the reflected signal. The device can calculate a time-dependent amplification of the reflected signal. This is needed because waves with a high transit time have a small signal amplitude.

The second displaying method is the B-Mode (brightness modulation). Here, the ultrasonic probe is swept evenly over the surface of the sample. The echo intensity of the reflected signal is displayed in a grayscale. The gray value of a pixel on the screen is a measure of the amplitude of the reflected signal.

3 Speed of Sound

In order to examine a material with ultrasound, knowledge of the material-dependent sound velocity is indispensable. The ultrasonic device allows this velocity to be determined by the transit time and the length of the body due to the reflection at boundary surfaces.

$$v_s = \frac{2nd}{\tau_n} \tag{1}$$

The distance is d, the transit time is τ_n, the order of the reflex is n. The distance d was determined with the help of a caliper and the running time by the time until

Material	Frequency[MHz]	Velocity[$\frac{\mathrm{Km}}{\mathrm{s}}$]
Polyacryl	1	(2.657 ± 0.055)
PVC	1	(2.302 ± 0.010)
PVC	2	(2.310 ± 0.023)
PVC	4	(2.336 ± 0.024)
Water	2	(1.483 ± 0.019)

Table 1: The table contains the experimentally determined sound velocities for the materials polyacrylic, PVC and water. Measurements were taken at different frequencies.

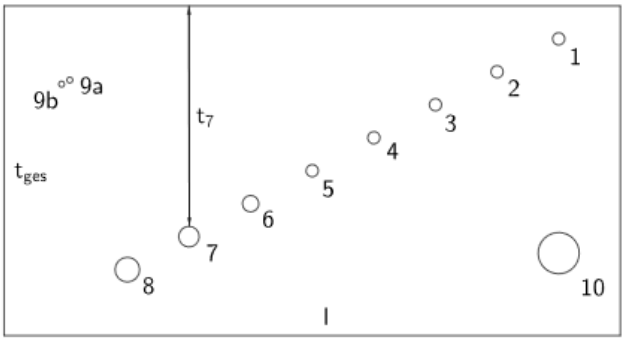

Fig. 1: This sketch shows the structure of the polyalcrylic block. The depth d was measured with a caliper gauge and ultrasonic waves. Taken from [1].

the rising edge of the reflex pulse. This was done for PVC and polyacryl as well as for water. The velocities for PVC don't agree within their errors with the literature value of $v = 2395\frac{\mathrm{m}}{\mathrm{s}}$ [6]. The reason for this is that PVC is a composition of different substances. Since the same proportion is not always used, deviations can occur. With this knowledge, our experimental data can be reconciled with the literature value. The velocity for water agrees within its error with the literature value of $v_{20^\circ} \approx 1482.9\frac{\mathrm{m}}{\mathrm{s}}$ [5]. For polyacryl the determined value agrees within its error with the literature value of $v \approx 2700\frac{\mathrm{m}}{\mathrm{s}}$ [4].

4 Resolution

The pulse width of the transmitted sound waves is finite. Two reflections whose distance is too small, can therefore not be separated, because the pulses would overlap. The full width at half maximum (FWHM) is suitable to quantify the resolution of the measurement. The results of all measurements agree in their errors. Furthermore, considering the errors of the measurements, it can be assumed that the precision of the measurement increases with increasing frequency. The vertical difference of the holes 9ab was determined with a caliper gauge to be $d_{9ab}^{cal} = (1,67 \pm 0,57)$ mm. The measured value with the all three probes agrees within its error with the literature specification. In addition, the proximity of the measured value with the 4 MHz probe and the measurement with the caliper reinforces the assumption of increased precision at higher frequencies.

Frequency[MHz]	FWHM[μs]	Distance 9ab[mm]
1	(1.250 ± 0.088)	(1.76 ± 0.28)
2	(1.88 ± 0.18)	(1.75 ± 0.33)
4	(0.900 ± 0.071)	(1.66 ± 0.19)

Table 2: The table contains the specific FWHM. Furthermore, the distance between t_{9a} and t_{9b} was determined via the signal time between the resolved peaks.

5 Absorption

In all examinations it could be observed that as soon as a second reflex is visible its intensity is significantly reduced and usually had to be amplified with the time gain control (TGC). The TGC allows the amplification to be increased with the runtime in order to counteract the damping. The decrease follows an exponential course. Furthermore, it could be observed that the damping is stronger at higher frequencies. This occurs because the atoms oscillate faster and the wave therefore scatters more. The resulting absorption coefficients are $\lambda_{2MHz} = (0.3243 \pm 0.0033)\frac{1}{cm}$ and $\lambda_{1MHz} = (0.2441 \pm 0.0256)\frac{1}{cm}$. Since the values of the different frequencies are not compatible with each other, they were given separately. The different values at different frequencies lead to the assumption that the absorption is frequency dependent. Taking into account the results of section 4, a compromise must be made between the damping and the resolution. Therefore, the further measurements were taken with the 2MHz probe.

6 Measurements on a polyacrylic block

In this part of the experiment, ultrasonic waves were used to determine the position (depth d_i) of holes in a polyacrylic block. The examined bock can be seen in figure 1. To get an understanding of how accurate ultrasonic examinations are, the hole depths d_i were first measured by hand using a caliper gauge. Furthermore, the total length of the polyacrylic block t_{ges} was also measured. It should be noted that this was somewhat

t_i	d_i−ultrasonic waves [mm]	d_i−caliper [mm]
t_1	(7.65 ± 0.16)	(7.03 ± 0.40)
t_2	(14.64 ± 0.30)	(14.94 ± 0.40)
t_3	(22.97 ± 0.47)	(23.10 ± 0.40)
t_4	(30.62 ± 0.63)	(31.10 ± 0.40)
t_5	(37.94 ± 0.78)	(39.06 ± 0.40)
t_6	(45.26 ± 0.93)	(46.55 ± 0.40)
t_7	(52.9 ± 1.1)	(54.22 ± 0.40)
t_8	(59.9 ± 1.2)	(61.50 ± 0.40)
t_{9a}	(17.64 ± 0.37)	(17.47 ± 0.40)
t_{9b}	(19.80 ± 0.41)	(19.14 ± 0.40)
t_{10}	(54.6 ± 1.1)	$55.41 \pm 0.40)$
t_{ges}	(78.2 ± 1.6)	(80.17 ± 0.40)

Table 3: Here are the results of the various depth measurements t listed. It can be seen that almost all values agree within their error.

more difficult than expected, so an error of 0.4 mm was estimated. The measured depths are listed in table 3.

An ultrasound probe with frequency $f = 2$ Mhz was placed on the upper side of the polyacrylic block. The transit time of the ultrasound wave was recorded using the A-mode. Then, together with the speed of sound in polyacrylic (see chapter "Speed of sound"), the depth of the holes can be calculated according to the equation 1. This calculated depths are also listed in table 3. When comparing the hole depths, it is noticeable that all depths t, except for t_1, agree within the limits of their error.

The reason for this is that an ultrasound gel has been distributed between the ultrasound probe and the block. This ultrasound gel ensures by means of its refractive index that the sound wave goes into the polyacrylic block. A transition would not be possible between the air and the polyacrylic block. However, the ultrasound gel leads to the fact that there is already a first reflection when penetrating the polyacrylic block. This is relatively close to the reflection of the first hole t_1. This leads to the reflection amplitudes merging into each other and can no longer be cleanly separated. This makes the ultrasound measurement somewhat less accurate compared to the measurement with the caliper.

6.1 Measurements in B-mode

The polyacrylic block was also measured using the B-mode. For this purpose, the ultrasonic probe was pulled over the top of the polyacrylic block at a constant speed. The figure shows two of these B-mode images. It can be seen that in the upper of the two images the ultrasound probe was not removed, while in the lower image vertical dropouts are noticeable. This means that the upper picture was done with more care than the lower. Moreover, is the upper one done slower, as the contrast between the dark and light spots is higher and have more data to evaluate. Nevertheless, the holes can be clearly seen in both images, which appear as white round areas. The top and bottom of the block are clearly visible as a horizontal line. Faint bright areas can be seen again below the underside. These are, of course, not holes, as the polyacrylic block has already ended there for a long time. These are the second-order reflections that occur after two reflections. They are only faintly

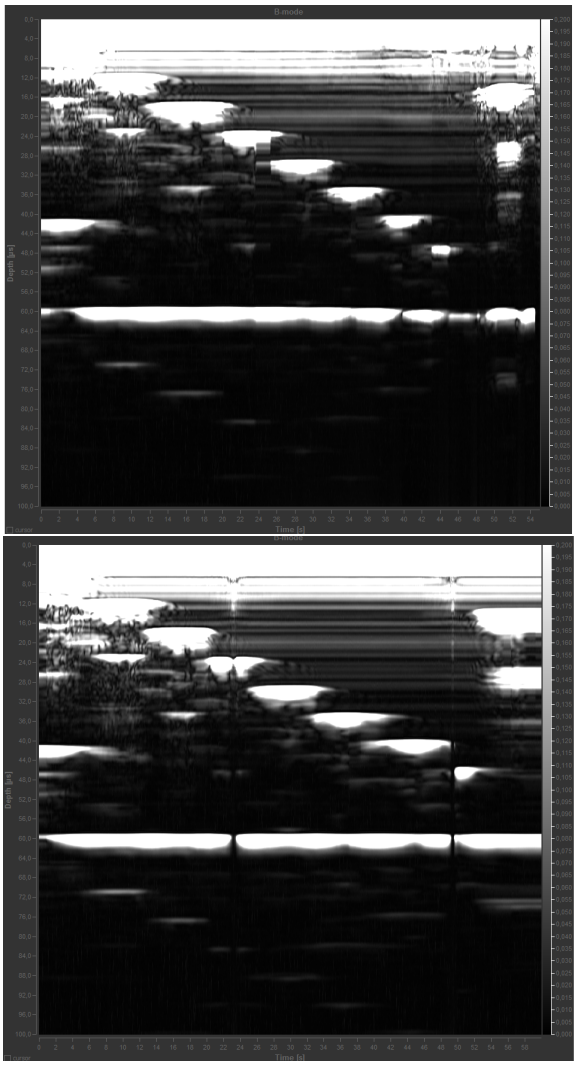

Fig. 2: This figures show the B-mode of polyacrylic block. The white areas are the holes. Top: The block was measured with a constant speed and the ultrasonic probe didn't disconnect form the top. Bottom: The block was measured by a higher speed and the ultrasonic probe was disconnected form the top twice.

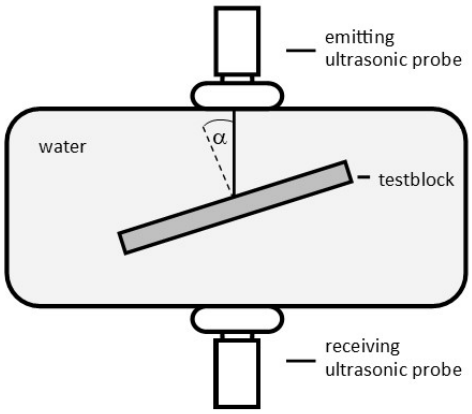

Fig. 3: The set-up for determining the speed of sound in copper and aluminum can be seen. The change in the transmitted signal was documented at the measured angle.

white because the intensity decreases significantly with longer running time (see chapter Absorption). The B-mode image is exactly mirrored to the figure 1. The first hole t_1 can't be differed clearly from the top of the block. The reasons therefore are stated in the former paragraph. The holes t_{9a} and t_{9b} can be differed as two separate bright spots on the right side end of the figure.

7 Speed of sound in solids

Sound propagates in solids in the form of longitudinal and transverse waves. Depending on the type of vibration, the atoms vibrate either in or perpendicular to the direction of propagation. In the following, the propagation characteristics of sound in aluminum and copper will be investigated. Because of the specific periodic order of atoms where the strength of their bounds is directional, both materials are anisotropic crystals. This results in different propagation speeds for the different wave modes. To measure these velocities a test block was placed in water at different angles between an emitting and a receiving ultrasonic sensor. This setup is shown schematically in fig. 3. Using a signal with a frequency of 2 MHz, the transmitted intensity was measured as a function of angle. When passing through a boundary between two different media ultrasonic waves behave according to Snell's law

$$\frac{\sin\alpha}{\sin\beta} = \frac{v_w}{v_s}, \tag{2}$$

where α and β give the angles of the incoming and the deflected wave (see fig. 3), v_w the speed of sound in water (see table 1) and v_s the speed of sound in the solid. At a certain angle α_T total reflection occurs . From this angle, no further reflexes can be observed. If exactly the angle α_T is hit, then $\beta = 90°$ applies to the deflected wave. Further, transmission is maximum for $\beta = 45°$. Thus both angles α and β are known for a measured peak. Assuming that these exact angles were found in the measurement, v_s can now be calculated for both the transverse and the longitudinal wave. The results are as listed in table 4 with errors based on a maximum error estimation. The error of α results from the reading accuracy of the angular scale and the zero point correction.

Material	Mode	$\alpha \pm 0.35$	β	$v\ [\frac{\text{km}}{\text{s}}]$
Cu	trans.	54.00°	90°	1.833 ± 0.032
Cu	long.	17.75°	90°	4.86 ± 0.16
Al	trans.	30.75°	90°	2.900 ± 0.068
Al	trans.	21.50°	45°	2.861 ± 0.083
Al	long.	13.50°	90°	6.35 ± 0.25

Table 4: This table shows the results of the determination of the velocity of sound for transverse (trans.) and longitudinal (long.) waves in copper (Cu) and aluminum (Al).

All values are within their error consistent with the theoretical values shown in table 5 in the appendix.

8 Summary

In these experiments, ultrasound measurements were made in PVC, polyacryl, aluminum, copper and water. Different properties of the bodies were investigated. The speed of sound was determined for different frequencies in PVC to $v_{1MHz} = (2.302 \pm 0.010)\ \frac{\text{Km}}{\text{s}}$, $v_{2MHz} = (2.310 \pm 0.023)\ \frac{\text{Km}}{\text{s}}$ and to $v_{4MHz} = (2.336 \pm 0.024)\ \frac{\text{Km}}{\text{s}}$. For polyacryl it was determined to be $v_{1MHz} = (2.657 \pm 0.055)\ \frac{\text{Km}}{\text{s}}$. The velocity in water is $v_{2MHz} = (1.483 \pm 0.019)\ \frac{\text{Km}}{\text{s}}$. All the experimentally determined speeds agree with their literature values. Furthermore, the resolution of different frequencies was determined. This is done via the full width at half maximum (FWHM) of the peaks. A higher resolution with increasing frequency could be determined. The frequency-dependent and distance-dependent absorption coefficient was determined for PVC. This is $\lambda_{1MHz} = (0.2441 \pm 0.0256)\ \frac{1}{\text{cm}}$ and $\lambda_{2MHz} = (0.3243 \pm 0.0033)\ \frac{1}{\text{cm}}$. A polyacrylic block with holes of different depths was measured with ultrasonic waves. It was found that ultrasonic measurement at a frequency of $f = 2$ MHz can determine the depth of the holes well. Lastly, the speed of sound was determined in copper and aluminum. Angle-dependent reflections and transmissions were recorded for this purpose. With copper and aluminum, both the longitudinal and the transverse waves are within the intended range.

References

[1] Praktikumsanleitung, Versuch C35 Messungen mit Ultraschall, last visited 22th January 2022

[2] CODATA recommended values of the fundamental physical constants: 2018, Tiesinga et al.,Reviews of modern physics, Volume 93, 2021

[3] Hoskins, Peter; Martin, Kevin; Thrush, Abigail: Introduction to B-Mode imaging, Cambridge University Press, 2010, last visited: January 23, 2022 https://assets.cambridge.org/97805217/57102/excerpt/9780521757102-excerpt.pdf

[4] Sound velocity of polyacrylic, GAMPT mbH, last access on January 25, 2022, https://www.gampt.de/kontakt-und-service/downloads-2/# (Katalog Ausbildung, p. 21)

[5] Eichler, Hans Joachim; Kronfeldt, Heinz-Detlef; Sahm, Jürgen: Das neue Physikalische Grundpraktikum, 3. Auflage, Springer, Berlin, 2016, p. 104

[6] Sound velocity of PVC, Olympus, last visited January 25, 2022, https://www.olympus-ims.com/de/ndt-tutorials/thickness-gauge/appendices-velocities/

9 Appendix

Mathematically, waves can be described by a set of differential equations

$$\rho \frac{\partial^2 u_i}{\partial t^2} = \sum_{jkl} c_{ijkl} \frac{\partial^2 u_l}{\partial x_j \partial x_k}, \tag{3}$$

with the postponement in x_i-direction u_i and the Young's modulus c. Assuming that the wave is propagating in the x-y-plane these equations can be solved and the theoretical angular dependent sound velocities can be obtained. The results are plotted in figure 4 Thereby one gets a velocity for the longitudinal mode as well as for the horizontal and the vertical transverse mode. The range between the extreme values of the respective velocities is shown in Table 5.

Material	Mode	theo. range of speeds $[\frac{\mathrm{km}}{\mathrm{s}}]$
copper	transverse	$1.62504 - 2.91082$
copper	longitudinal	$4.35011 - 4.9755$
aluminum	transverse	$2.93298 - 3.24413$
aluminum	longitudinal	$6.31333 - 6.46376$

Table 5: This table contains the numerical evaluation of the figures 4 of the speed of sound in copper and aluminum.

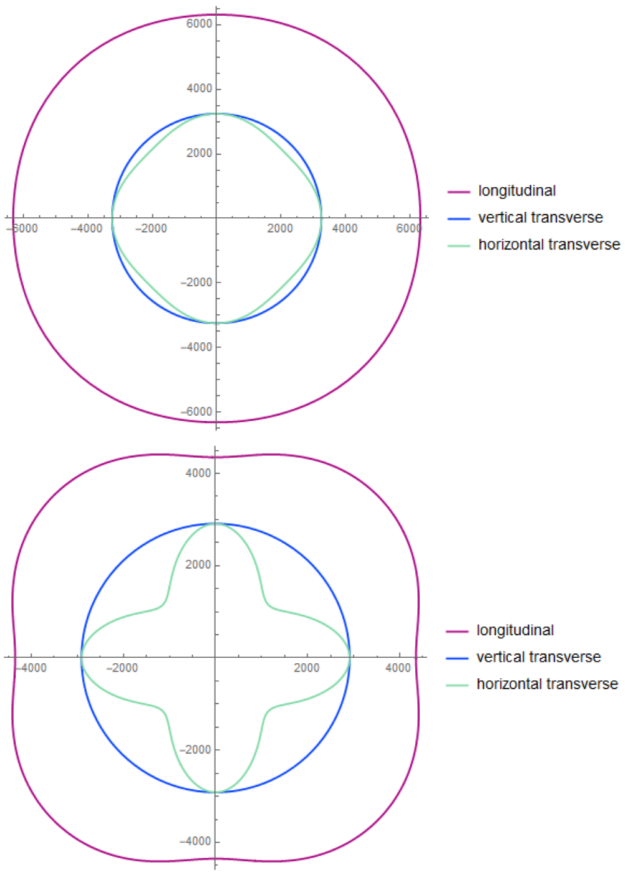

Fig. 4: These pictures show the theoretical speed of sound of the transverse waves (vertical and horizontal) and the longitudinal waves as a function of the measured angle. The upper one represents the speed in aluminum, the lower one in copper.